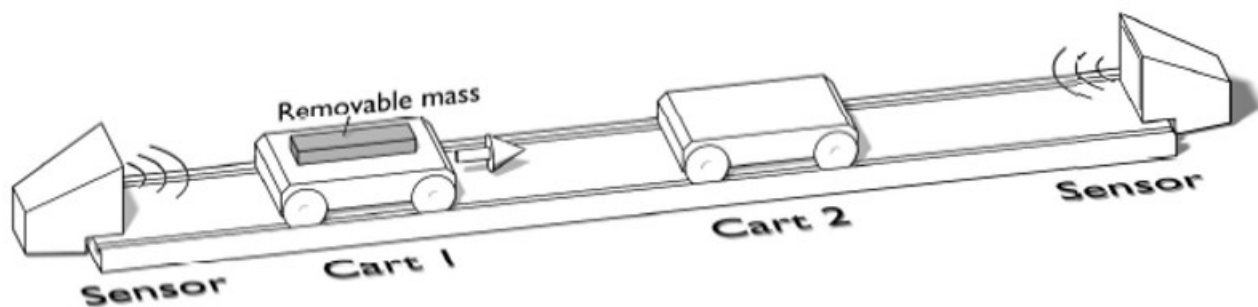


Lab 8

Collisions and Conservation Laws



Physical Models: Conservation of Linear Momentum, Conservation of Mechanical Energy

Analysis Tools: t-score, propagation of uncertainty

Experimental Systems: PASCO Collisions Carts, Weights

Equipment: Capstone motion detectors

Safety Concerns: Don't let the carts collide with the motion detectors or fly off the tracks.

Introduction

The goal of lab this week will revolve around investigating collisions. Collisions can be very complex, which can make tracking motions of objects undergoing collisions moment to moment very difficult. Because of this, it's often useful to analyze collisions in terms of *conservation laws*. A conservation law states that regardless of what happens to a system, no matter how complicated, some quantity will remain the same. However, conservation laws usually require certain conditions to be met for them to hold true.

In the Prelab Activity you learned about two candidate conserved quantities and the conditions under which we can expect them to hold in a two-body collision:

Linear Momentum $\vec{P} = m_1\vec{v}_1 + m_2\vec{v}_2$ Conserved when there are no external forces.

Mechanical Energy $E = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$ Conserved when there are no dissipative forces

Our goal in this lab will be to investigate whether or not these quantities are actually conserved as expected in a real collision process.

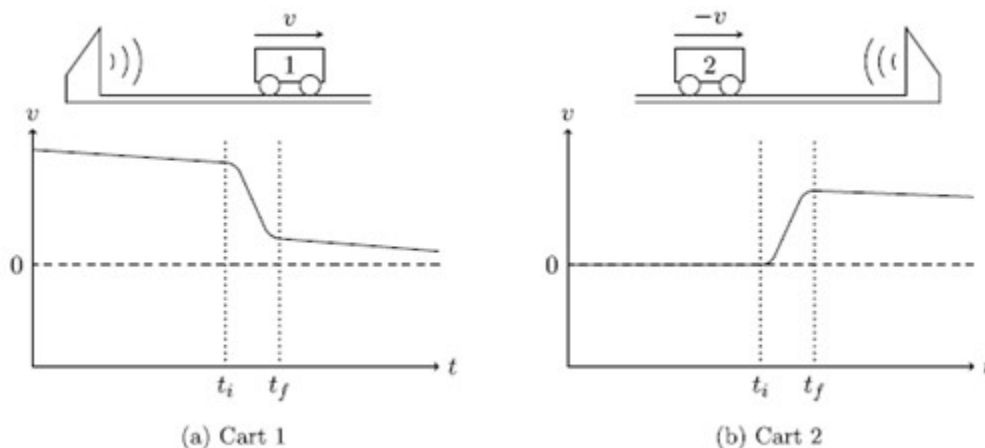
Part 1. Conservation of Energy

First let's take a look at an actual collision. You have two carts, red and blue, which will serve as the objects undergoing a collision. Set them on the track (at rest) any way you choose for now, we'll get more systematic later.

Open "CollisionSensor.cap". Make sure the PASCO motion detectors are setup correctly on the track as shown, and that they are plugged into the PASCO interface in slots 1 and 3.

Quick Check: Once everything is setup let's take a look at a collision! Turn the detectors on using the PASCO start button. You should see graphs of the velocity vs. time for each cart. Lightly push the red cart into the blue cart so that they collide, as in the picture on the first page of the lab.

Let the collision proceed, then stop the carts and turn off the detectors. You should get results like this:



Notice the plots are very complicated, especially near the collisions! It'd be quite difficult to analyze these curves with Newton's second law $F=ma$. But if we focus on conservation laws and only compare the initial and final states, things simplify.

Designing Your Experiment: Now, write out a plan for a high-precision measurement to determine whether or not energy conservation holds for a given type of collision. You have two sets of options for the collisions:

- You can either leave the blue cart free to move (Free Target) or hold it fixed in place by flipping it upside down on the track and having a group member hold it (Fixed Target).
- You can either have the cart's magnets facing each other (Elastic) or the Velcro (Inelastic).

You need to try all four combinations at least once each. You must find at least one collision for which mechanical energy is not conserved.

You'll need to decide:

- What do you expect to happen for each collision based on what was presented in the prelab activity, the descriptions on the first page of the lab, and the results of your quick check?

- You will need to think about a range of masses and distances to check the model over. As we discovered in the last couple of weeks, just because a model works in a small range doesn't mean it works universally!
- You will need to think about how to minimize uncertainty given how it propagates (to begin with, you may want to just try a number of points and see how things shake out, but you may want to become more systematic once you get a feel for it. Record choices as you go!)
- How will you determine a measurement for the initial and final velocities and their associated uncertainties?
- How will you decide if the quantity is conserved or not?
- How does uncertainty propagate given your method? You can refer to the Prelab for assistance.

Some things to keep in mind:

- The sensors report velocities as + or - using their own definition of direction. You may find it more useful to define your own coordinate system (e.g. rightward is always + and leftward -) for the whole system and convert the sensor readings accordingly.
- The cart masses can be treated as 275 ± 5 grams.
- The weights can be treated as 250 ± 5 grams.
- Your TA has a precision scale should you decide it's necessary.
- The Prelab gave a walkthrough of the necessary propagation of uncertainty for this lab, including the final resulting formulas you'll need.

Check in with Another Group: Briefly discuss your plans with another group. They may provide you with additional ideas and feedback. Note one similarity and one difference between your approaches.

Conducting Your Experiment: Go ahead and give it a shot. You can change your method around while you're at it, you don't need to treat your previous decisions as a contract.

Follow Up: Select one of the cases where you found energy was not conserved. Develop a hypothesis for why the collision did not conserve energy, and *test this* by collecting data on at least one more collision. Consider increasing or decreasing the variable that resulted in mechanical energy loss.

Forming a Conclusion: Does your data support the prediction regarding the conservation law given the kind of collision you investigated? If not, what are some possible reasons this might be? If so, how could you improve the precision of the experiment?

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare? Note two similarities and/or differences between what you found, and use this to inform your proposals for future iterations.

Post Your Results on the Board: Your TA should have provided a table like this one on the board:

	Free Target	Fixed Target
Elastic		
Inelastic		

Add your results to the appropriate box. You can write simply “E Conserved” or “E not Conserved”, whatever your results indicate.

Part 2. Conservation of Momentum

In this part we’ll repeat Part 1, but for momentum rather than energy. Remember that momentum is a vector, so the signs of the velocities matter. You may need to interpret the values provided by the sensors very carefully because of this! Which direction is positive? Negative? Does it depend on the sensor?

Quick Check: Try out a collision without collecting any data. Does it look like momentum is conserved? Make some predictions about which collisions will or won’t conserve momentum.

Designing Your Experiment: Design an experiment following the same requirements as Part 1, but this time for momentum conservation. You’re welcome to reuse data from collisions in Part 1 if you wish, but make sure to include at least 1 completely new trial for your “Follow Up” (see below).

Conducting Your Experiment: Go ahead and give it a shot. You can change your method around while you’re at it, you don’t need to treat your previous decisions as a contract.

Follow Up: Select one of the cases where you found momentum was not conserved for the system of carts. Develop a hypothesis for why the collision did not conserve momentum, and *test this* by collecting data on at least one more collision.

Forming a Conclusion: Does your data support the prediction regarding the conservation law given the kind of collision you investigated? If not, what are some possible reasons this might be? If so, how could you improve the precision of the experiment?

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare? Note two similarities and/or differences between what you found, and use this to inform your proposals for future iterations.

Post Your Results on the Board: Your TA should have provided a table like this one on the board:

	Free Target	Fixed Target
Elastic		
Inelastic		

Add your results to the appropriate box. You can write simply “P Conserved” or “P not Conserved”, whatever your results indicate.

Writing Up Your Lab Notes for Submission

Write up your notes for Part 1 and Part 2 in an organized way according to the rubric provided.